

Microdam Impoundments Provide Suitable Habitat for Larvae of Malaria Vectors: An Observational Study in Western Kenya.

PMID: 29462354 | Indexed in PubMed

Impoundments formed by microdams in rural areas of Africa are important sources of water for people, but they provide potential larval habitats for *Anopheles* (Diptera: Culicidae) mosquitoes that are vectors of malaria. To study this association, the perimeters of 31 microdam impoundments in western Kenya were sampled for *Anopheles* larvae in three zones (patches of floating and emergent vegetation, shorelines of open water, and aggregations of cattle hoofprints) across dry and rainy seasons. Of 3,169 larvae collected, most (86.8%) were collected in the rainy season. Of 2,403 larvae successfully reared to fourth instar or adult, nine species were identified; most (80.2%) were *Anopheles arabiensis* Patton, sampled from hoofprint zones in the rainy season. Other species collected were *Anopheles coustani* Laveran, *Anopheles gambiae* s.s. Giles, *Anopheles funestus* Giles, and *Anopheles rivulorum* Leeson, *Anopheles pharoensis* Theobald, *Anopheles squamosus* Theobald, *Anopheles rufipes* (Gough), and *Anopheles ardensis* (Theobald). Larvae of *An. funestus* were uncommon (1.5%) in both dry and rainy seasons and were confined to vegetated zones, suggesting that microdam impoundments are not primary habitats for this important vector species, although microdams may provide a dry season refuge habitat for malaria vectors, contributing to population persistence through the dry season. In this study, microdam impoundments clearly provided habitat for the malaria vector *An. arabiensis* in the rainy season, most of which was within the shallow apron side of the impoundments where people brought cattle for watering, resulting in compacted soil with aggregations of water-filled hoofprints. This observation suggests a potential conflict between public health concerns about malaria and people's need for stable and reliable sources of water.

The prevalence of *Anopheles* (Diptera: Culicidae) mosquitoes in Sekong Province, Lao PDR in relation to malaria transmission.

PMID: 12791058 | Indexed in PubMed

A longitudinal study was conducted on the prevalence of *Anopheles* in three malaria endemic villages in Sekong province, in the southern region of Lao PDR, from August 2000 to October 2001. All night, human landing collections took place in August and October 2000 and April and October 2001, and blood smears were taken for malaria parasites during the same period. Mosquitoes were tested for sporozoite antigen using enzyme-linked immunosorbent assay. In August 2000 (wet season) and April 2001 (dry season) the ovaries of the mosquitoes were examined for parity. A total of 16 species of *Anopheles* were caught in the study sites of which *An. dirus* A, *An. maculatus* sl and *An. jeyporiensis* were positive for sporozoites. The entomological inoculation rate (EIR) ranged from 0.06 to 0.25. There was a good correlation between EIR and vectorial capacity in the wet season, especially in Pai Mai where the prevalence of malaria was also high during the wet seasons (11.8 and 10.53). *An. dirus* A showed ambivalence in their choice of feeding as approximately 50% attacked man indoors and an equal proportion outdoors. *An. dirus* A was the main vector in Pai Mai. The parous rate did not significantly differ between the wet and dry season, although it was higher in the dry season. In Takaio the parasite prevalence ranged from 8.7% (dry season) to 37.1% (wet season) and *An. jeyporiensis* was the vector, and the risk of infection was 0.85 in the dry season while 0.99 in the wet season. In Toumgno *An. maculatus* sl was the vector and infection was found only in August and October 2000. However, malaria prevalence ranged from 9.69 to 20.4% and was equally high in the dry season. Cattle were also present close to the houses in all the villages

and this might be a contributory factor in the prevalence of malaria.

Adult Anopheles Mosquito Distribution at a Low and High Malaria Transmission Site in Tanzania.

PMID: 35047638 | Indexed in PubMed

Malaria parasites are only transmitted by female mosquitoes of the genus *Anopheles*; hence, the disease's distribution is linked to that of the vector mosquitoes. As such, the goal of this study was to find out the spatial and temporal distribution of *Anopheles* mosquito adults in the research sites. This was a repeated cross-sectional ecological study that took place in Morogoro and Dodoma, Tanzania. Vacuum aspiration was used to collect mosquitoes both outside and inside human dwellings. All mosquito-related data was collected and entered into appropriate data collection forms. Female mosquitoes were recognized morphologically using Gillies and Coetzee morphological criteria, followed by PCR. In total, about 2742 *Anopheles* mosquitoes with an average collection of 18.21 ± 1.12 per day were collected outside human houses of which 1717 (10.51 ± 1.17) and 1025 (8.42 ± 1.41) were collected from Morogoro and Dodoma, respectively. Of the captured mosquitoes, 89.0%, 10.0%, and 1.0% were recognized as *Anopheles arabiensis*, *Anopheles gambiae* s.s., and *Anopheles quadrianulatus*, respectively. The distribution varied significantly with seasons, whereby 302 (4.72 ± 1.04) and 2440 (12.96 ± 1.52) mosquitoes were captured in the cold-dry and warm-wet season, respectively ($p < 0.0001$). Of the captured mosquitoes, 42.33%, 16.33%, 14.96%, and 4.27 were found on the ceiling, stored junks, verandas, and barks/tree, respectively. In malaria-endemic countries, vector control forms an important component of the malaria control efforts. This study found significant variation of *Anopheles* mosquito abundance in time and space with *Anopheles arabiensis* being the most predominant malaria vector. This signifies the need to introduce mosquito control methods that will target the less anthropophilic *Anopheles arabiensis* or the immature aquatic stages. The study further found that underbeds, store room/piled bags, and undisturbed curtains were the most preferred resting places by mosquitoes signifying to be the most effective strategic sites for spraying insecticides during the implementation of indoor residual spraying (IRS).

The Plasmodium bottleneck: malaria parasite losses in the mosquito vector.

PMID: 25185005 | Indexed in PubMed

Nearly one million people are killed every year by the malaria parasite *Plasmodium*. Although the disease-causing forms of the parasite exist only in the human blood, mosquitoes of the genus *Anopheles* are the obligate vector for transmission. Here, we review the parasite life cycle in the vector and highlight the human and mosquito contributions that limit malaria parasite development in the mosquito host. We address parasite killing in its mosquito host and bottlenecks in parasite numbers that might guide intervention strategies to prevent transmission.

Standard Membrane Feeding Assay for the Detection of Plasmodium falciparum Infection in Anopheles Mosquito Vectors.

PMID: 35635457 | Indexed in PubMed

Malaria remains one of the most devastating diseases worldwide and, to date, the African region is still responsible for 94% of all cases worldwide. This parasitic disease requires a protozoan parasite, an *Anopheles* mosquito vector, and a vertebrate host. The *Anopheles* genus comprises more than 500 species, of which 60 are known as vectors of the parasite. The *Plasmodium* parasite genus consists of 250 species, and 48 of these are involved in disease transmission. Furthermore, the *Plasmodium falciparum* parasite has contributed toward an estimated 99.7% of malaria cases in sub-Saharan Africa in recent years. Gametocytes form part of the sexual stage of the parasite and are ingested by the female mosquito upon feeding on an infected human host. Further development of the parasite within the mosquito is enhanced by favorable environmental conditions in the midgut of the mosquito. Here, the fusion of the female and male gametes takes place, and the motile ookinetes originate. The ookinetes enter the midgut epithelium of the mosquito, and mature ookinetes form oocysts, which, in turn, produce motile sporozoites. These sporozoites migrate to the mosquito's salivary glands and are injected as a mosquito takes a blood meal. For drug discovery purposes, mosquitoes were artificially infected with gametocyte-infected blood in the standard membrane feeding assay (SMFA). To detect infection within the mosquito and/or to assess the efficacy of antimalarial compounds, the midguts of the female mosquitoes were removed post infection and were stained with mercurochrome. This method was used to enhance the visual detection of oocysts under the microscope for the accurate determination of oocyst prevalence and intensity.

A mating-induced reproductive gene promotes *Anopheles* tolerance to *Plasmodium falciparum* infection.

PMID: 33347501 | Indexed in PubMed

Anopheles mosquitoes have transmitted *Plasmodium* parasites for millions of years, yet it remains unclear whether they suffer fitness costs to infection. Here we report that the fecundity of virgin and mated females of two important vectors-*Anopheles gambiae* and *Anopheles stephensi*-is not affected by infection with *Plasmodium falciparum*, demonstrating that these human malaria parasites do not inflict this reproductive cost on their natural mosquito hosts. Additionally, parasite development is not impacted by mating status. However, in field studies using different *P. falciparum* isolates in *Anopheles coluzzii*, we find that Mating-Induced Stimulator of Oogenesis (MISO), a female reproductive gene strongly induced after mating by the sexual transfer of the steroid hormone 20-hydroxyecdysone (20E), protects females from incurring fecundity costs to infection. MISO-silenced females produce fewer eggs as they become increasingly infected with *P. falciparum*, while parasite development is not impacted by this gene silencing. Interestingly, previous work had shown that sexual transfer of 20E has specifically evolved in *Cellia* species of the *Anopheles* genus, driving the co-adaptation of MISO. Our data therefore suggest that evolution of male-female sexual interactions may have promoted *Anopheles* tolerance to *P. falciparum* infection in the *Cellia* subgenus, which comprises the most important malaria vectors.

Malaria infectivity of xanthurenic acid-deficient anopheline mosquitoes produced by TALEN-mediated targeted mutagenesis.

PMID: 29349579 | Indexed in PubMed

Anopheline mosquitoes are major vectors of malaria parasites. When the gametocytes of the malaria parasite are transferred from a vertebrate to mosquitoes, they differentiate into gametes, and are fertilized in the midguts of mosquitoes. Xanthurenic acid (XA), a waste product of the ommochrome synthesis pathway, has been shown to induce exflagellation during microgametogenesis in vitro; however, it currently remains unclear whether endogenous XA affects the infectivity of anopheline mosquitoes to malaria parasites in vivo due to the lack of appropriate experimental systems such as a XA-deficient line. In the present study, we produced a XA-deficient line in *Anopheles stephensi* using transcription activator-like effector nuclease (TALEN)-mediated gene targeting (knockout) of the kynurenine 3-monooxygenase (*kmo*) gene, which encodes an enzyme that participates in the ommochrome synthesis pathway. The knockout of *kmo* resulted in the absence of XA, and oocyst formation was inhibited in the midguts of these XA-deficient mosquitoes, which, in turn, reduced sporozoite numbers in their salivary glands. These results suggest that endogenous XA stimulates exflagellation, and enhances the infectivity of anopheline mosquitoes to malaria parasites in vivo. The XA-deficient line of the anopheline mosquito provides a useful system for analyzing and understanding the associated factors of malaria gametogenesis in the mosquito midgut.

Molecular characterization of anopheline mosquitoes from the goat malaria-endemic areas of Thailand.

PMID: 36598082 | Indexed in PubMed

Despite the fact that over a 100 anopheline mosquito species have been identified as human malaria vectors, little is known about ungulate malaria vectors. Consequently, we focused on investigating the bionomics and genetic characterizations of anopheline mosquitoes in goat malaria-endemic regions. We also attempted to screen for ungulate malaria potential vectors. A total of 1019 female anopheline mosquitoes were collected from six goat farms in four provinces of Thailand from 2020 to 2021. Mosquitoes were morphologically identified and subsequently confirmed using the mitochondrial DNA barcoding region-cytochrome oxidase c subunit I (MtDNA-COI), mitochondrial DNA-cytochrome c oxidase subunit II (MtDNA-COII), and ribosomal DNA internal transcribed spacer 2 (rDNA-ITS2) sequences. The current study reveals the genetic characteristics and distribution of nine mosquito species within the *Anopheles* and *Cellia* subgenera. Four dominant species, including *Anopheles peditaeniatatus*, *Anopheles subpictus*, *Anopheles vagus*, and *Anopheles aconitus* exhibited significant intraspecific gene flow within their corresponding species. Although malaria parasites were not found in 126 mosquito pools, meaning more investigation is necessary, the current study adds to the existing DNA barcoding data collection and improves the current understanding of the genetic structure and distribution of anopheline mosquito species, which could be useful for effective control of mosquito-borne diseases.

Simplified Plasmodium falciparum membrane feeding assay using small Petri dishes and gel warmers.

PMID: 39710720 | Indexed in PubMed

Studies on *Plasmodium falciparum* transmission require blood-feeding infectious gametocytes to mosquitoes using standard membrane-feeding assays (SMFAs). SMFAs are routinely performed using electric heating coils or glass membrane feeders connected to a circulatory water bath using tubing and

clamps. Each of these approaches is expensive and requires a complex setup, hence restricting the number of assays that can be performed simultaneously. Furthermore, existing methods cannot be easily applied in low-resource field settings. This study presents a low-cost and simplified method for feeding mosquitoes with an infectious blood meal using 35 mm Petri dishes where temperature is maintained by using reusable gel warmers. The intensity and prevalence of infection in mosquitoes (*Anopheles stephensi* and *Anopheles gambiae*) fed via a Petri dish overlaid with gel warmers were comparable to mosquitoes fed using standard glass membrane feeders. The methodology described here can be easily applied in low-resource and field settings due to its low cost, ease of set up, and use of easily available supplies, such as Petri dishes, and reusable gel warmers. We believe a wide range of laboratories can easily adapt this method for *P. falciparum* transmission studies.

Use of Synthetic Membranes and Bovine Blood in Artificial Feeding Systems of Mosquitoes Improves Feeding Rates: Results from a Systematic Review and Meta-Analysis.

PMID: 39689365 | Indexed in PubMed

Artificial blood feeding of mosquitoes is essential for their rearing in insectaries as well as for mosquito-borne pathogen transmission experiments. We conducted a systematic review and meta-analysis of the different artificial feeding systems available for mosquitoes to synthesize evidence regarding their efficacy in terms of feeding rates and fecundity. PubMed, Scopus, Web of Science, and Google Scholar were systematically searched to retrieve 1,822 experimental studies assessing the efficacy of artificial feeding systems. After assessing eligibility and risk of bias, 25 studies were included in the final analysis. Studies were reviewed and meta-analysis was conducted using random-effects model. The primary outcomes were feeding rates and fecundity of mosquitoes belonging to genera *Aedes*, *Culex*, and *Anopheles* for the different systems. Subgroup analyses with respect to membrane, blood source, and mosquito genera were conducted. Sensitivity analysis was done to assess heterogeneity. The overall pooled estimate of feeding rate of mosquitoes and the average number of eggs laid per female mosquito using artificial blood-feeding systems were 72% (66-77%) and 71.9 (56.68-87.12), respectively. Results from this systematic review are suggestive of the advantages of novel techniques, such as Digital Thermo Mosquito Blood Feeder and 3D-printed feeders. The study provides evidence for improved feeding rates of mosquitoes in systems using bovine blood and synthetic membranes such as latex.